

Dear Editor,

We would like to submit the manuscript entitled “DSN: Energy-Efficient EMG Signal Classification Method Enabling Real-Time Monitoring for Edge Healthcare”, which we wish to be considered for publication in Neural Networks.

Surface electromyography (sEMG) signal classification is a key technology for wearable edge healthcare, but it is hindered by high computational complexity and energy consumption—critical limitations for battery-constrained devices. While Spiking Neural Networks (SNNs) excel at modeling the sparse, time-varying nature of sEMG signals, and Hyperdimensional Computing (HDC) enables robust pattern recognition with limited data, their standalone use fails to balance efficiency and accuracy.

Motivated by these challenges, we propose DSN, a hybrid framework integrating SNNs and HDC to address the efficiency-accuracy tradeoff in edge sEMG monitoring. More concretely, our contributions are summarized as follows:

1. We develop a synergistic SNN-HDC fusion architecture, where the SNN uses randomized untrained weights for lightweight spatiotemporal feature extraction and the HDC maps features to high-dimensional space for robust classification, achieving 93.87% average accuracy in three-class fatigue classification. It has the ability to maintaining over 90% accuracy using only 20% of the training data.

2. The framework realizes ultra-low energy consumption, with energy consumption $42.42\times$ lower than GRU and $26.87\times$ lower than LSTM, supporting long-term continuous monitoring on wearables.

3. We provide rigorous theoretical validation, including Lipschitz stability analysis of SNN, HDC operation formalization, and three theorems proving noise robustness, Cluster-HV representational fidelity, and hardware error resilience—ensuring the framework’s reliability.

I am the corresponding author and my address and other information is as follows,

Address:

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology,
Wuhan, 430074,
P.R.China

E-mail: zgzen@hust.edu.cn

Thank you very much for consideration!

Sincerely Yours,

Zhigang Zeng